

Depth of the Minimal-Width Coding Memorizer

An exponential barrier, a precision-uniform optimality theorem, and a high-dimensional speedup for
arXiv:2411.08735

Automated proof attempt
agent_lane_03 model: GPT5.6

11 August 2026

Status: substantial partial result, likely valid, pending human verification. For all finite-parameter piecewise-affine LReLU and stepped-LReLU classes in the source, worst-case fixed-width depth is necessarily exponential in the input coding parameter K . If the depth cap is required to be independent of output precision M , the source’s $O(2^{d_x K})$ memorizer depth is optimal up to width constants. Conversely, a post-source ReLU memorization theorem gives depth about the square root of that bound at the source’s exact minimal width whenever that width is at least 40. Low minimal widths and networks allowed to exploit FLOOR remain open.

Source question

The exact excerpt is Remark 79 and Table 3 on printed page 52 of the current source paper (the following image is a direct crop of that page).

Remark 79 *Our coding scheme constructions in Theorems 10, 11, and 18 require a number of layers that grows exponentially with the approximation parameters K and M . Table 3 summarizes the layer counts for each component across different activation classes. We emphasize that these are upper bounds achieved by our constructions; whether more depth-efficient implementations exist at minimal width remains an open question.*

Component	$\mathcal{F}_+, \mathcal{F}_\pm, \mathcal{F}_{+,s}, \mathcal{F}_{\pm,s}$	with FLOOR	$\{\sigma_\alpha\}, \{\sigma_{-\alpha}, \sigma_\alpha\}$
Encoder	$\mathcal{O}(2^K)$	2	$\mathcal{O}(2^K \cdot g_1(\epsilon))$
Memorizer	$\mathcal{O}(2^{d_x K})$	$\mathcal{O}(2^{d_x K})$	$\mathcal{O}(2^{d_x K} \cdot g_2(\epsilon))$
Decoder	$\mathcal{O}(d_y 2^M)$	$\mathcal{O}(d_y 2^M)$	$\mathcal{O}(d_y 2^M \cdot g_3(\epsilon))$

Table 3: Layer counts for our coding scheme constructions. For single-parameter activations, $g_i(\epsilon) \rightarrow \infty$ as $\epsilon \rightarrow 0$ reflects additional depth from LReLU approximations.

For the activation classes $\mathcal{F}_+$, $\mathcal{F}_\pm$, $\mathcal{F}_{+,s}$, and $\mathcal{F}_{\pm,s}$, our constructions directly build the required piecewise linear functions:

- *The encoder from Lemmas 67 and 69 approximates the quantizer q_K , which has 2^K linear pieces. By Lemma 61, this requires $\mathcal{O}(2^K)$ layers.*

The source uses $C_n = \{j2^{-n} : 0 \leq j < 2^n\}$. Its encoder bijects the grid $\mathcal{C}_K^{d_x}$ onto $Z_K := \mathcal{C}_{d_x K}$, so

$$N := |Z_K| = 2^{d_x K}.$$

For a continuous target $f : [0, 1]^{d_x} \rightarrow [0, 1]^{d_y}$, its memorizer maps the corresponding point of Z_K to the encoded M -bit quantization of f at the grid point. The source realizes this scalar interpolation with depth $O(N)$, independently of M .

Call an activation family *tame piecewise affine* if its graph is a semialgebraic family with a bounded number of affine pieces and at most two real activation parameters per unit. This includes all of the source's families

$$\mathcal{F}_+, \mathcal{F}_\pm, \mathcal{F}_{+,s}, \mathcal{F}_{\pm,s}, \mathcal{F}_{\alpha,s}, \mathcal{F}_{\pm\alpha,s}, \mathcal{F}_{+,1}^*, \mathcal{F}_{\pm,1}^*,$$

and all their fixed-slope subfamilies. It does not include FLOOR.

Theorem 1 (Worst-case depth trichotomy). *Fix scalar network width $w \geq 1$, $d_x, d_y, K \geq 1$, and a tame piecewise-affine source activation family. Let $N = 2^{d_x K}$.*

- (a) *If depth at most L suffices to realize the exact coding memorizer for every continuous target already at one output bit ($M = 1$), then*

$$L \geq c \frac{\sqrt{N}}{(w+1)^{3/2}} = c \frac{2^{d_x K/2}}{(w+1)^{3/2}},$$

where $c > 0$ is universal. The same conclusion holds if exact matching is replaced by uniform error strictly below $1/4$ on the coding grid.

- (b) *Suppose one depth cap $L = L(K)$, independent of M , realizes the exact memorizer for every $M \geq 1$ and every continuous target. Then*

$$L \geq \frac{N}{w(w+3)}.$$

Thus the source's precision-uniform $O(N)$ memorizer is order-optimal for fixed w .

- (c) *There is also a complementary upper bound. For any fixed $\alpha \in (0, 1)$, arbitrary coding data admit an exact σ_α -LReLU memorizer of width at most 40 and depth*

$$O(\sqrt{NB} + B), \quad B := 1 + d_x K + d_y M.$$

Consequently, whenever the minimal width in the source is at least 40, this is an implementation at that minimal width. For $d_x K + d_y M = O(d_x K)$, parts (a) and (c) match in their exponential order $2^{d_x K/2}$, up to polynomial factors in K .

Proof Intuition

The target function is arbitrary, so its values on the finite input grid can carry arbitrary labels. Exact coding at one bit therefore forces the network class to shatter all N encoded grid points. A width- w , depth- L piecewise-affine network has only $O(w^2 L)$ weights and $O(wL)$ nonlinear units, so the sharp general VC upper bound is $O(w^3 L^2)$. This gives the square-root exponential barrier.

Requiring the same depth bound for every M is much stronger: dyadic label vectors of all precisions are dense in the full N -cube. The set of evaluation vectors of bounded-depth tame networks is semialgebraic and has dimension at most its $O(w^2 L)$ real parameters. Density forces dimension N , hence a linear-in- N depth bound.

The upper bound uses a March 2026 width–depth memorization theorem, then converts every ReLU layer exactly into a fixed-LReLU layer at twice the width.

Proof

Enumerate the input grid as $G_K = \{x_1, \dots, x_N\} = \mathcal{C}_K^{d_x}$, and write $z_j = \mathbf{e}_K(x_j) \in Z_K$. The z_j 's are distinct. We first record why the target class really supplies arbitrary grid data.

Lemma 2 (Arbitrary finite coding data). *For every $M \geq 1$ and every vector $(a_1, \dots, a_N) \in \mathcal{C}_M^N$, there is a continuous $f : [0, 1]^{d_x} \rightarrow [0, 1]^{d_y}$ whose coding memorizer satisfies*

$$\mathbf{m}_{K,M,f}(z_j) = a_j \quad (1 \leq j \leq N).$$

Proof. Choose pairwise disjoint neighborhoods of the finitely many x_j 's and continuous functions $0 \leq \eta_j \leq 1$, supported in those neighborhoods, with $\eta_j(x_j) = 1$. Set

$$f(x) = \left(\sum_{j=1}^N a_j \eta_j(x), 0, \dots, 0 \right).$$

Disjointness of the supports keeps the first coordinate in $[0, 1]$. At x_j , its M -bit quantization is exactly a_j , and encoding a vector whose only nonzero entry is its first entry leaves that entry unchanged. This is the asserted identity. $\square$

For part (a), apply Lemma 2 with labels $a_j \in \{0, 1/2\} \subset \mathcal{C}_1$. Thresholding the scalar network output at $1/4$ realizes every binary labeling of Z_K , so Z_K is shattered.

To count capacity, first eliminate the variable activation parameters. For all real β ,

$$\sigma_\beta(t) = \text{ReLU}(t) - \beta \text{ReLU}(-t), \quad \rho_{\beta,s}(t) = \text{ReLU}(t) - \beta \text{ReLU}(-t) + s \text{STEP}(t).$$

Thus each source unit can be simulated by at most three units with fixed piecewise-polynomial activations, with the affine layer after it performing the displayed linear combination. A width- w , depth- L source network is consequently contained in a fixed-activation network class with at most

$$W_* \leq C_1(w+1)^2 L \quad \text{parameters}, \quad U_* \leq C_2 w L \quad \text{nonlinear units}.$$

Bartlett–Harvey–Liaw–Mehrabian's general piecewise-polynomial bound [2, Theorem 8] gives

$$\text{VCdim} \leq C_3 W_* U_* \leq C_4 (w+1)^3 L^2.$$

Since the VC dimension is at least N , rearranging proves (a). If the grid error is less than $1/4$, thresholding still recovers the binary labels, so the same proof applies.

For part (b), let $E_{w,L} \subset \mathbb{R}^N$ be the set of all evaluation vectors $(\phi(z_1), \dots, \phi(z_N))$ from widths at most w and depths at most L . For each fixed depth and width profile, the graph of evaluation in the affine weights, biases, and activation parameters is semialgebraic. Tarski–Seidenberg therefore makes $E_{w,L}$ a finite union of semialgebraic sets. A depth- n network has at most

$$nw(w+1) + 2w(n-1) \leq nw(w+3)$$

real parameters. Dimension cannot increase under a semialgebraic map, and semialgebraic closure preserves dimension; hence

$$\dim \overline{E_{w,L}} \leq Lw(w+3).$$

The hypothesis and Lemma 2 give $\mathcal{C}_M^N \subset E_{w,L}$ for every M . The union of these dyadic grids is dense in $[0, 1]^N$, so $[0, 1]^N \subset \overline{E_{w,L}}$. Its dimension is N , proving $N \leq Lw(w+3)$ and part (b).

For part (c), apply Yang–Yang's Theorem 2.1 [3] to the N scalar inputs $z_j \in [0, 1]$, whose separation is $1/N$, and to the $2^{d_y M}$ possible coded labels (rescale their integer indices at the output). In their explicit formula choose the bit-block parameter at least both $\log R$ and $\log(2^{d_y M})$, where here $R = 10N^3\sqrt{\pi}$, and choose the sample-block size as the ceiling of the square-root balancing value. Their displayed width is then at most 20, while their depth is

$$O\left(\sqrt{N(1 + \log N + d_y M)} + 1 + \log N + d_y M\right).$$

Since $\log N = d_x K \log 2$, this is the asserted bound with B .

Finally, a width- r ReLU layer has an exact width- $2r$ realization using one fixed positive LReLU, because

$$\text{ReLU}(t) = \frac{\sigma_\alpha(t) + \alpha \sigma_\alpha(-t)}{1 - \alpha^2}.$$

At each layer the affine map produces z and $-z$ in parallel, and the next affine map performs the displayed combination. This doubles width, does not increase depth, and yields width 40. The activation σ_α belongs to every relevant variable positive/stepped source family (take step size zero), completing (c). $\square$

What is closed, and what remains open

For tame LReLU families, part (a) gives a definitive qualitative answer to the source’s final suggestion that exponential scaling “may be unavoidable”: no worst-case implementation with depth subexponential in K exists at any fixed width. Part (b) is sharp for the source’s stronger, implicit precision-uniform memorizer specification. Part (c) shows that the linear-in- N construction can nevertheless be improved at fixed precision, and at exact minimal width in dimensions where that width is at least 40.

Three boundaries prevent classification as a full answer. First, the exact fixed-precision complexity at minimal widths below 40, especially scalar width one or two, is not settled. Second, FLOOR is not semialgebraic with a bounded number of pieces, so neither capacity proof applies to classes that may use it. This is substantive: with FLOOR and identity, a width-two, constant-depth network can extract the i -th bit of a real parameter on geometrically spaced inputs, giving infinite VC dimension. Third, the decoder’s exponential dependence on M is not closed here.

The durable attempt note records eight focused upgrade attempts, including the failed width-one serialization and FLOOR-grid bit-extraction routes.

Reproducibility and literature boundary

The script `code/verify_depth_bounds.py` checks the exact ReLU-to-LReLU identity over rational test data, the parameter-count inequality, and the constant-depth FLOOR bit-extraction obstruction. These checks do not replace the VC-dimension or semialgebraic-dimension proofs.

The source was revised in December 2025. Yang–Yang was submitted in March 2026 and therefore postdates the source; its theorem is used only for part (c). Bartlett et al. supplies the pre-existing capacity theorem used in part (a). The precision-uniform closure argument and the application to this specific coding question are proved above. No duplicate packet or ledger entry was found in the run indexes. This is a bounded novelty check, not an exhaustive priority claim.

References

- [1] D. Rochau, R. Chan, and H. Gottschalk, *New advances in universal approximation with neural networks of minimal width*, arXiv:2411.08735, current version December 2025. Included as `source_paper.pdf`; the exact two-page source excerpt is also included as `source_excerpt.pdf`.
- [2] P. L. Bartlett, N. Harvey, C. Liaw, and A. Mehrabian, *Nearly-tight VC-dimension and pseudodimension bounds for piecewise linear neural networks*, JMLR 20 (2019), no. 63, 1–17; arXiv:1703.02930. Included under `references/`.
- [3] X. Yang and Y. Yang, *Memorization capacity of deep ReLU neural networks characterized by width and depth*, arXiv:2603.09589 (10 March 2026). Included under `references/`.